Expanded Joint Interceptor Assignment for Cooperative Multi-Target Tracking

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Abstract

This report studies interceptor assignment conflicts in a mobile wireless sensor network (WSN) for cooperative multi-target tracking. In Naeimeh's previous work [1], interceptor conflicts were resolved through a MinMax objective that attempts to minimize the worst team cost among the targets involved in the same conflict-resolution step. This formulation provides a fairness-oriented baseline, but it does not directly optimize the total assignment cost. In addition, when a later handover request takes over a sensor that has already been committed to another target, the previously served target may lose that committed sensor, thereby reducing its probability of successful interception.

In this report, the proposed method is referred to as an expanded joint assignment approach. Here, "expanded" means that the joint assignment scope includes not only the current calling targets, but also any previously served target whose committed interceptor is contested by a later handover request. The method differs from the previous approach in three ways. First, it includes both the current calling targets and the previously served targets whose committed interceptors are contested in the same joint assignment problem. Second, it uses a two-stage assignment objective that first maximizes the number of filled interceptor slots and then minimizes the total sensor-to-target assignment cost among assignments with the same number of filled slots. Third, it replaces non-overlapping candidate team comparison with an event-local sensor-to-target assignment over all involved targets and eligible candidate sensors. Under ideal communication assumptions in simulation, the resulting formulation retains the distributed execution logic of the original DES framework while providing a different mechanism for resolving interceptor assignment conflicts. The communication requirements and implications for physical-platform implementation are discussed later in the report.

1. Introduction

Mobile wireless sensor networks (WSNs) provide a flexible platform for cooperative multi-target tracking because mobile sensors can reposition themselves in response to target motion and local tracking needs. In Naeimeh's previous work [1], this capability was organized through a distributed Discrete Event System (DES) framework in which sensors coordinate proactive handover through local bid computation, handover-request broadcasting, and Finite State Machine (FSM)-based execution. During operation, an active tracker issues a handover request when target loss is predicted, and available candidate sensors participate in an auction-like bidding process to determine which sensors should serve as interceptors. The selected interceptors then move toward the predicted interception region to support tracking continuity before the current tracker loses the target. This distributed handover framework provides the system baseline for the present report.

This report focuses on a narrow but important issue within that framework: interceptor assignment conflict resolution. In the DES framework, interceptor assignment conflicts arise when multiple handover requests compete for the same candidate sensors or when a later handover request attempts to use a sensor already committed to another target. In the previous method, conflicts among handover requests within the same conflict-resolution step were handled through a MinMax objective that attempts to minimize the worst team cost among the involved targets. This formulation provides a fairness-oriented baseline and helps reduce the likelihood that any target receives a poor interceptor configuration. However, because the MinMax objective optimizes the worst-case team cost rather than the total assignment cost, it does not necessarily produce the most efficient overall assignment. In addition, although the previous work already provides a reasonable DES-based mechanism for resolving conflicts that occur within an assignment interval, another limitation remains: a later handover request can reassign a sensor that has already been committed to another target without simultaneously reconsidering the target that loses this interceptor. When that happens, the previously served target loses one of its committed interceptors, which can reduce its probability of successful interception.

This report formulates committed-interceptor conflict resolution as an expanded joint assignment problem. Compared with the previous MinMax-based method, the proposed approach differs in three main ways. First, it expands the joint assignment scope beyond the current calling targets by including any previously served target whose committed interceptor is contested by a later handover request. Second, it adopts a two-stage assignment objective that first maximizes the number of filled interceptor slots across the involved targets and then minimizes the total assignment cost among assignments that achieve the same number of filled slots. Third, it replaces the previous comparison of non-overlapping candidate teams with a constrained sensor-to-target assignment model that jointly enforces each target's interceptor capacity and each sensor's one-target-at-a-time constraint. Under ideal communication assumptions in simulation, this revised formulation preserves the distributed execution logic of the original DES framework while providing a different mechanism for resolving committed-interceptor conflicts. The communication requirements and implications for physical-platform implementation are discussed later in the report.

2. Related Work

Multi-target tracking in wireless sensor networks (WSNs) has evolved from reactive detection-based strategies to predictive and coordinated frameworks. Early approaches primarily relied on passive tracking, where sensors detect targets within range and reacquire them after loss through opportunistic encounters. While simple, such methods may suffer from frequent tracking interruptions when targets move quickly or when sensor coverage is sparse.

To address these limitations, predictive tracking frameworks have been developed to anticipate tracking loss and enable proactive coordination. The DES-based coordination framework proposed in [1] integrates Extended Kalman Filter (EKF)-based loss prediction, guidance-based sensor repositioning, and distributed bidding for sensor allocation. In that framework, sensors operate using a finite-state-machine architecture and transition between tracking and repositioning roles, enabling proactive handover before target loss occurs. This framework provides the baseline system architecture for the present report.

Task allocation in multi-robot and sensor networks is commonly handled through auction-based or bidding-based mechanisms because they allow agents to evaluate task suitability locally. These mechanisms are suitable for distributed systems, but conflicts may arise when multiple targets require support from the same limited set of candidate sensors. Therefore, the assignment objective plays an important role in determining how limited sensor resources are allocated. A MinMax objective provides a fairness-oriented solution by reducing the worst target-level team cost, while total-cost-based objectives prioritize lower combined assignment cost under the same feasibility constraints.

In the DES framework of [1], interceptor conflict resolution is handled using a MinMax objective when multiple targets compete for candidate sensors within the same assignment process. This provides a meaningful fairness-oriented baseline. However, the original formulation mainly addresses conflicts within the active assignment procedure and does not explicitly include the later DES condition in which a new handover request contests a sensor that has already been committed as an interceptor for another target. In this case, the previously served target may be affected even though it is not part of the current calling-target set.

This report builds upon [1] by expanding the assignment scope to include targets affected by committed-interceptor conflicts. When a committed interceptor is contested by a later handover request, the target that may lose that interceptor is added to the expanded joint assignment problem before the assignment is solved. The report then applies an event-local two-stage assignment objective over the involved targets and eligible candidate sensors.

Unlike the MinMax objective, which emphasizes worst-team-cost fairness, the proposed two-stage objective first maximizes the number of filled interceptor slots and then minimizes the total assignment cost among feasible assignments with the same number of filled slots.

3. System Model and Problem Formulation

3.1 System Model and Assumptions

This section models the interceptor assignment problem. Consider a WSN with N mobile sensors that cooperatively track up to K targets. Each target can be assigned up to two active tracking sensors, and up to m interceptor sensors when a potential target loss is predicted.

Each sensor, $i \in \mathcal{N} = \{1, \ldots, N\}$, has a home position, $h_i \in R^2$ and a current position $s_i(t) \in R^2$. Each target, $k \in K = \{1, \ldots, K\}$, , has state $x_k(t) = [p_k(t), v_k(t)]$, where $p_k(t)$ is the target position and $v_k(t)$ is the target velocity. Sensors have identical capabilities: detection radius r_d, maximum velocity v_s and communication range r_c. Target detection occurs when $\|s_i(t) - p_k(t)\| \leq r_d$. Upon detection, sensor i obtains a noisy position measurement of target k, $z_{ik}(t) = p_k(t) + n(t)$ where $n(t)$ is zero-mean measurement noise with covariance R.

In the underlying DES framework, proactive handover is triggered when an active tracker predicts that its target may soon be lost. The tracker then broadcasts a handover request to available candidate sensors. The candidate sensors participate in a bidding process, in which each candidate computes a bid metric representing its suitability for the handover task. The selected candidates are assigned as interceptors and are dispatched toward the predicted interception region to support tracking continuity before target loss occurs. Sensors assigned to execute such a handover request are referred to as interceptors in the remainder of this report.

Throughout the report, calling targets are targets that initiate a handover request and enter the interceptor assignment process. Candidate sensors are eligible sensors that may bid for those targets, while active trackers are the sensors currently responsible for direct target tracking and are excluded from interceptor bidding during the event. An assignment event consists of the broadcast, bidding, and selection stages initiated by one or more handover requests.

The system assumptions used in the remainder of the report are as follows: the workspace is planar, sensors are homogeneous, assignments are event-triggered, and each sensor can serve at most one target at a time within the current assignment event. Because different targets may enter the assignment process at different times, a later handover request may compete with sensors that have already been committed to another target. The proposed method addresses this case by expanding the joint assignment scope before solving the assignment problem.

3.2 Problem Formulation and Bidding Cost Function

When one or more targets issue handover requests, the system defines the set of targets included in the expanded joint assignment and the set of candidate sensors. The target set includes the current calling targets and any previously served targets whose committed interceptors may be reassigned due to the current conflict-resolution event.

For each calling target, one of its active trackers first predicts an interception point based on the current EKF target estimate and the predicted tracker-loss point. This information is then broadcast to the network. Each eligible candidate sensor evaluates a bidding cost for that target. A lower bid indicates that the sensor is a more suitable interceptor candidate.

Following the formulation in [1], the bidding cost is defined as:

$$c(i,k) = w_1\, P\_home(i) + w_2\, P\_spatial(i,k) + w_3\, P\_temporal(i,k)\; + w_4\, P\_uncertainty(i,k)$$

- $P\_home(i)$ penalizes the displacement of a sensor from its home location.
- $P\_spatial(i,k)$ penalizes the distance from the sensor to the target intercept point.
- $P\_temporal(i,k)$ penalizes poor arrival timing relative to the target motion.
- $P\_uncertainty(i,k)$ penalizes low shared confidence in the target estimate.

Let $x(i,k)$ indicate whether sensor i is assigned to target k, then we can have:

$$x(i,k) = \begin{cases} 1, & if\ sensor\ is\ assigned\ to\ target\ k \\ 0, & otherwise \end{cases}$$

Let $\mathcal{N}_c$ denote the set of candidate sensors and $\mathcal{K}_a$ denote the set of ~~involved~~ targets in the expanded joint assignment ~~problem.~~ First, the first stage maximizes the total number of interceptor slots filled across the targets in the assignment scope:

$$max \sum_{k\in\mathcal{K}_a}\sum_{i\in\mathcal{N}_c} x(i,k)$$

The second stage then minimizes the total assignment cost over all feasible assignments that achieve the maximum number of filled ~~interceptor~~ slots ~~determined in~~ the first stage.:

$$min \sum_{k\in\mathcal{K}_a}\sum_{i\in\mathcal{N}_c} c(i,k)x(i,k)$$

The assignment is subject to the following constraints:

$$\sum_{k\in\mathcal{K}_a} x(i,k) \le 1, \qquad \forall\, i \in \mathcal{N}_c$$

$$\sum_{i\in\mathcal{N}_c} x(i,k) \le m, \qquad \forall\, k \in \mathcal{K}_a$$

$$x(i,k) \in \{0,1\}, \qquad \forall\, i \in \mathcal{N}_c, \forall\, k \in \mathcal{K}_a$$

The first constraint ensures that each candidate sensor can be assigned to at most one target in the current assignment event. The second constraint ensures that each involved target receives no more than its allowed number of interceptors, where m is the maximum number of interceptors per target.

4. Methodology

4.1 Conflict Detection and Expanded Joint Assignment Scope

The interceptor assignment workflow is activated when an active tracker predicts that a target may soon be lost and initiates a handover request. Within the DES process, a conflict is detected when multiple handover requests compete for one or more of the same candidate sensors. This can occur either because their assignment intervals overlap or because a later handover request attempts to use a sensor that has already been committed to another target.

In [1], conflicts are identified at the event level by checking whether the same candidate sensor appears in more than one active request set. This report extends the conflict condition to include the case where a new handover request attempts to employ a sensor that has already broadcast the commitment message to another target. This extension ensures that the previously served target is also included in the expanded joint assignment problem when one of its committed interceptors is contested by a later handover request, since losing an interceptor may increase its risk of target loss.

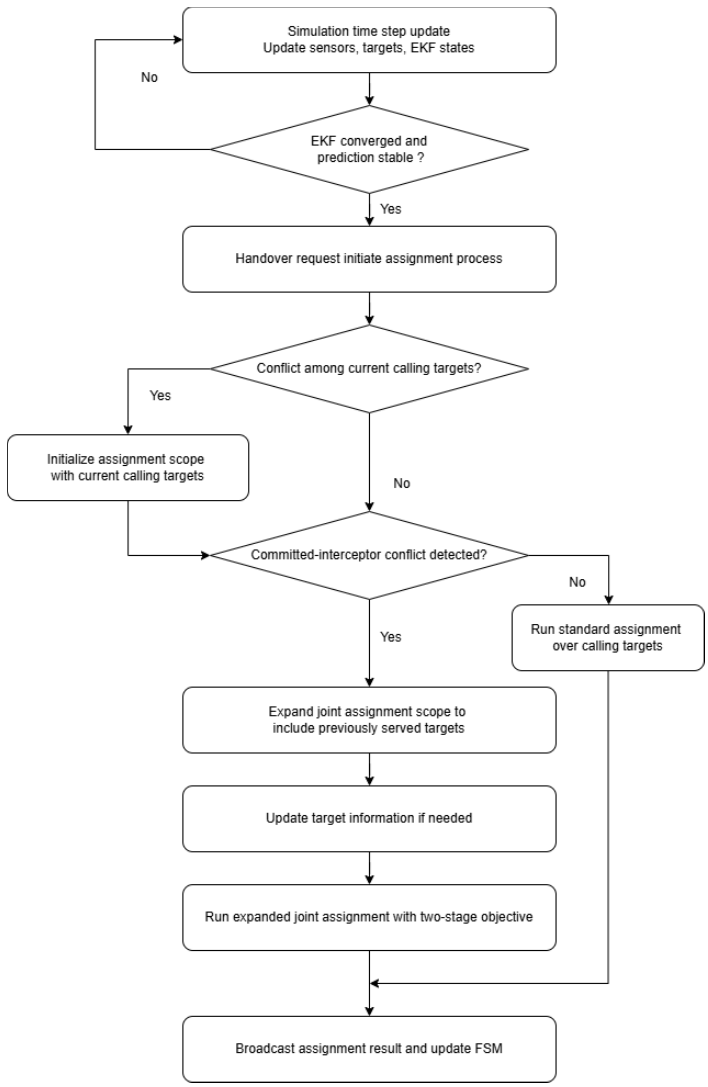

Figure 1: Joint interceptor assignment workflow in the DES framework.

To address this issue, the proposed method expands the joint assignment scope when necessary. The joint assignment scope initially consists of the current calling targets in the assignment event. If the conflict involves only these current handover requests, no further scope expansion is needed. The scope is expanded only when a committed-interceptor overlap is detected. In that case, the target that may lose its committed interceptor is added to the joint assignment problem. The assignment problem is then solved once over the expanded joint assignment scope, so that any target whose committed interceptor is contested is considered within the same assignment step.

4.2 Two-Stage Assignment Objective

The assignment problem introduced in Section 4.1 assigns eligible candidate sensors to interceptor slots for multiple involved targets under capacity constraints. The proposed method ensures that each sensor can be assigned to at most one target within the current assignment event, while each involved target receives no more than its allowed number of interceptors. The method uses a two-stage assignment objective: it first maximizes the number of filled interceptor slots across the involved targets and then minimizes the total assignment cost among assignments with the same number of filled slots.

Table 1 illustrates a simplified example involving two targets, T_1 and T_2, and five eligible candidate sensors: S_{10}, S_{13}, S_{14}, S_{18}, and S_{20}. In this example, a later handover request from T_2 attempts to employ S_{14}, while S_{14} has already been committed as an interceptor for T_1. Following the expanded joint assignment defined in Section 4.1, both T_1 and T_2 are included in the same assignment problem.

Candidate Sensor	Bidding Cost for Target 2	Bidding Cost for Target 1
S_{10}	0.377	0.300
S_{13}	0.306	0.304
S_{14}	0.219	0.193
S_{18}	0.327	0.397
S_{20}	0.285	0.377

Table 1: Example bidding-cost matrix for two targets.

In this example, each target can receive up to two interceptor sensors. Therefore, the assignment problem can fill up to four interceptor slots in total: two for T_1 and two for T_2. Each candidate sensor, however, can be assigned to at most one target. The algorithm therefore cannot simply select the two lowest-cost sensors for each target independently, because the same sensor may be preferred by more than one target. Instead, the assignment must be solved jointly over the involved targets while enforcing non-overlapping sensor use.

Table 2 compares the assignments produced by three different selection rules. The greedy assignment processes the targets sequentially and is therefore order dependent. The MinMax assignment minimizes the worst target-level team cost, which provides a fairness-oriented solution. The proposed two-stage assignment objective first fills the maximum feasible number of interceptor slots and then selects the feasible assignment with the lowest total cost.

Algorithm	Target 2	Target 2 Cost	Target 1	Target 1 Cost	Total	Worst Team Cost
Greedy [T2, T1]	S_{14}, S_{20}	0.504	S_{10}, S_{13}	0.604	1.108	0.604
MinMax	S_{13}, S_{20}	0.591	S_{14}, S_{18}	0.590	1.182	0.591
Two-Stage	S_{13}, S_{20}	0.591	S_{10}, S_{14}	0.493	1.085	0.591

Table 2: Assignment comparison under greedy, MinMax, and two-stage assignment methods.

The greedy method assigns S_{14} and S_{20} to T_2 first, leaving S_{10} and S_{13} for T_1. This produces a total assignment cost of 1.108. Although this result is feasible, it depends on the order in which targets are processed and does not solve the joint assignment problem optimally.

The MinMax method considers both targets but optimizes a different objective. It selects the assignment that minimizes the worst target-level team cost. In this example, MinMax assigns S_{13} and S_{20} to T_2, and S_{14} and S_{18} to T_1, producing a worst team cost of 0.591 and a total cost of 1.181. This reflects the fairness-oriented behavior of the MinMax objective.

The proposed two-stage method selects S_{13} and S_{20} for T_2, and S_{10} and S_{14} for T_1. This assignment fills all four interceptor slots and achieves the lowest total cost among the feasible assignments shown in Table 2. Therefore, within the involved targets and eligible candidate sensors of the current assignment event, the proposed method obtains the total-cost optimal assignment after satisfying the maximum-slot objective.

4.3 *Objective Trade-off Between Fairness and Total Assignment Cost Minimization*

The MinMax method and the proposed two-stage assignment method differ fundamentally in their optimization objectives. The MinMax method minimizes the worst-case target-level team cost, leading to a fairness-oriented assignment that helps prevent any involved target from receiving a significantly inferior interceptor configuration. This objective is meaningful when balanced tracking support across targets is the primary system priority.

In contrast, the proposed two-stage assignment method first maximizes the number of filled interceptor slots across the involved targets and then minimizes the total assignment cost while preserving the maximum number of filled interceptor slots obtained in the first stage. This objective prioritizes assignment coverage and total assignment cost minimization rather than worst-case fairness. Therefore, the two methods may produce different assignments under the same bidding-cost matrix: MinMax may accept a higher total assignment cost to reduce the worst target-level cost, while the proposed method may accept a less balanced distribution of target-level costs if doing so reduces the total assignment cost after the filled-slot maximization step is satisfied. In this report, the two-stage assignment method is adopted because the goal is to resolve interceptor assignment conflicts by jointly considering all targets included in the expanded joint assignment and enforcing non-overlapping sensor use within the current assignment event.

4.4 *Feasibility as a Distributed System*

The proposed two-stage assignment method preserves the distributed DES architecture by avoiding a ~~permanent~~ coordinator or centralized bid collector. The method changes the assignment rule, but it does not change the sensor-level execution logic of the original framework from [1]. Each sensor still operates according to its own FSM, computes its own bid locally, and updates its own state based on the assignment result. The optimality of the method refers to the event-local assignment problem over the involved targets and eligible candidate sensors, rather than centralized control over the entire WSN.

When an active tracker predicts that interceptor support is required, it broadcasts a handover request. This request is propagated through the communication network, and any non-active-tracking sensor that receives the request before the bidding deadline and satisfies the feasibility conditions is treated as an eligible candidate. Each candidate sensor computes its own bid cost locally according to the same bidding function. Therefore, the proposed method does not require any sensor to compute costs on behalf of other sensors.

After receiving the request, candidate sensors broadcast their bid values during a fixed bidding window, denoted by Δt_{bid}. This window allows bid messages to propagate through the participating sensor group before the selection step is executed. In the simulation, Δt_{bid} can be set to one DES time step, or to a fraction of a time step, such as half a time step, depending on the desired trade-off between communication completeness and response speed. At the end of the bidding window, each participating sensor reconstructs the same event-local assignment problem using the bid information received before the deadline and applies the same deterministic two-stage assignment rule. The rule is identical for all participating sensors and enforces the same capacity constraints: each target can receive at most two interceptors, and each sensor can be assigned to at most one target.

Under ideal communication, all participating sensors receive the same bid information before the deadline. In that case, they reconstruct the same assignment problem and obtain the same final assignment without requiring a central sensor to collect bids or broadcast the decision. If a sensor determines from the reconstructed assignment that it has been selected as an interceptor, it transitions to the interceptor state and broadcasts a commitment message indicating that it has accepted the assignment. If the sensor is not selected, it remains idle for later handover requests. In this way, bid computation, assignment reconstruction, commitment, and state transition are all performed locally.

The fixed bidding window also provides a practical mechanism for handling communication delay. Bids that arrive before the selection deadline are included in the assignment calculation, while late bids are excluded from the current assignment event. This deadline-based rule is consistent with the real-time nature of the DES framework. A sensor whose bid cannot reach the participating group before the deadline cannot be safely used for the current handover decision, even if its motion-based bid cost is low. Therefore, communication reachability is treated as part of assignment feasibility. Future implementations may incorporate communication delay or link reliability directly into the bid cost, but in the present formulation this constraint is handled through the bidding deadline.

The total delay of the assignment process includes more than the computation time of the two-stage assignment rule. It also includes the request propagation time, the bidding window, local assignment computation and commitment broadcast:

$$\Delta t = \Delta t_{\text{request}} + \Delta t_{\text{bid}} + \Delta t_{\text{local\_assignment}} + \Delta t_{\text{commit}}$$

Therefore, real-time feasibility must be evaluated using both computation and communication costs. In the current simulation, the method follows the same ideal communication assumption used in [1], so the assignment results are meaningful for algorithmic comparison. However, deployment on an embedded robotic platform would require measuring or bounding bid propagation delay, packet loss, commitment confirmation time, and the probability of inconsistent assignment reconstruction. Thus, the proposed method remains distributed at the sensor-execution level, but its practical feasibility depends on whether the communication network can support the required bidding and commitment exchange within the DES selection window.

5. Simulated Experiments

5.1 Experimental Setup

The experiments are conducted using the simulation framework developed in [1]. The same DES architecture, sensor models, EKF-based state estimation, bidding-cost formulation, and target motion configurations are adopted to ensure a fair comparison between the original MinMax-based method and the proposed expanded joint assignment approach with a two-stage assignment objective. The main difference between the two versions lies in how interceptor assignment conflicts are handled once handover requests compete for the same candidate sensors or when a committed interceptor is contested by a later handover request.

The wireless sensor network consists of 25 mobile sensors arranged in a 5 × 5 hexagonal grid with an inter-node spacing $d = 12$ units. Sensors have a fixed detection range $r_d = 1.5$ units, communication range $r_c = 18$ units, which enables each sensor to coordinate with first and second-tier neighbors. In all experiments, the target speed is set higher than the maximum sensor speed, so target loss can occur naturally, and proactive handover is required to maintain tracking continuity. The hexagonal geometry provides efficient coverage while maintaining structured communication patterns for distributed coordination.

Because the sensor behaviors, target motion, EKF settings, and bidding-cost computation are kept the same in all experiments, any differences in assignment outcomes are caused by the assignment strategy itself. Specifically, the experiments compare the original MinMax-based conflict-resolution method with the proposed expanded joint assignment method, which expands the joint assignment scope when a committed-interceptor conflict occurs and solves the resulting assignment problem using a two-stage objective. Sections 5.2 and 5.3 use a two-target scenario to demonstrate the committed-interceptor conflict and the effect of expanding the joint assignment scope, while Section 5.4 uses a three-target example to compare the assignment-cost outcomes under MinMax and the proposed two-stage assignment method.

5.2 Committed-Interceptor Conflict in the Baseline Method

A two-target scenario is first used to illustrate the committed-interceptor conflict in the baseline method. This example focuses on the case where a later handover request affects an existing interceptor assignment that was already committed to another target.

At $t = 39.2$s, Target 1 selected S_{10} and S_{14} as its interceptors. S_{14} was therefore committed to Target 1 and started moving toward the predicted interception region for Target 1, as shown in Figure 2.

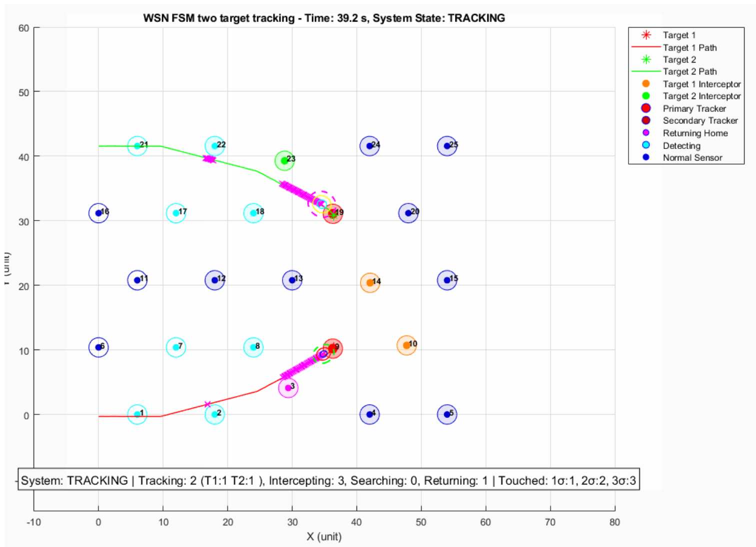

Figure 2: Baseline snapshot t=39.4s, $S_{10}$and S_{14} are committed to Target 1.

However, at $\mathrm{t} = 40.0$s, Target 2 entered the selection stage after issuing a later handover request. During this selection, Target 2 selected S_{20} and S_{14}. Since S_{14} had already been committed to Target 1, this created a committed-interceptor conflict. As shown in Figure 3, Target 2 took over S_{14}, and Target 1 lost one of its committed interceptors.

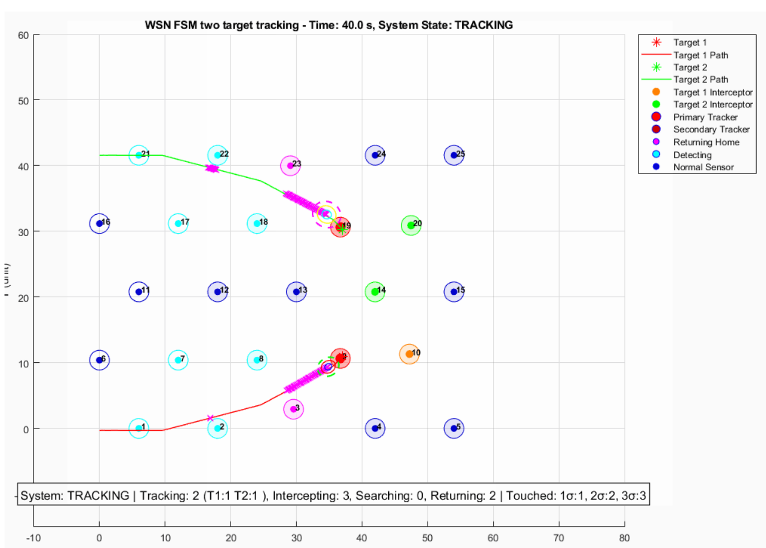

Figure 3: Baseline V45 snapshot at t=40.0s, Target 2 takes over S_{14} from Target 1.

This case shows that the baseline method does not include the previously served target in the assignment problem when one of its committed interceptors is contested by a later handover request. Although Target 1 and Target 2 were not handled within the same assignment procedure, the later handover request from Target 2 still affected the existing interceptor assignment of Target 1. This motivates the joint assignment scope expansion mechanism introduced in Section 4.1.

5.3 Effect of the Expanded Joint Assignment Scope

The proposed method addresses the committed-interceptor conflict by expanding the joint assignment scope before solving the assignment problem. When the later handover request from Target 2 attempts to use S_{14}, the system detects that S_{14} has already been committed to Target 1. Therefore, Target 1 is added to the joint assignment scope together with the current calling target, Target 2.

The assignment problem is then solved once over the expanded joint assignment and the eligible candidate sensors. This prevents the later handover request from directly overwriting the existing interceptor assignment of Target 1. Instead, both targets are considered within the same assignment step, and the two-stage assignment objective selects a feasible non-overlapping sensor-to-target allocation.

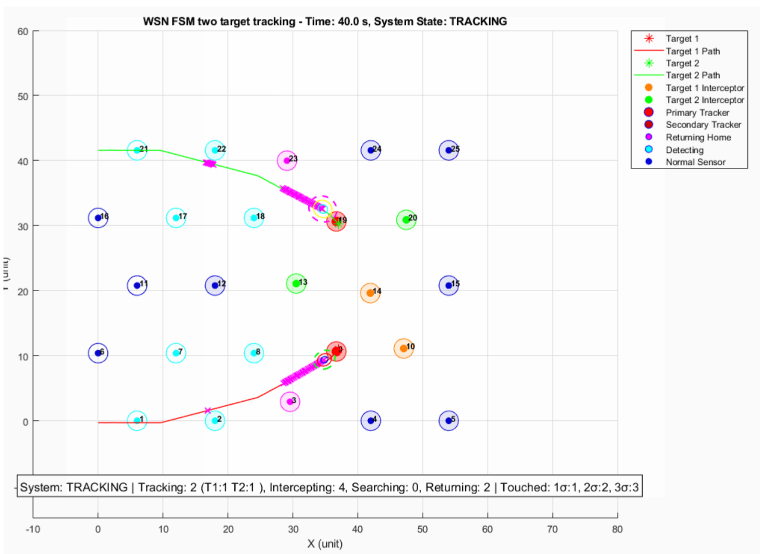

Figure 4: Proposed expanded joint assignment at $t = 40.0s$.

Figure 4 shows the result after applying the proposed expanded joint assignment approach. Target 2 no longer takes S_{14} away from Target 1. Instead, Target 2 selects S_{13} and S_{20}, while Target 1 retains S_{14} as one of its committed interceptors. This confirms that the proposed method includes the previously served target in the assignment problem when its committed interceptor is contested by a later handover request.

5.4 Three-Target Event: Separating Assignment Scope from Assignment Objective

This section uses a three-target event to separate two effects: the expanded joint assignment scope and assignment-objective selection. The original MinMax-based method without the expanded joint assignment scope is discussed as a scope limitation, because it would not form the full $[T_3, T_2, T_1]$ assignment problem as a single event-level instance. Therefore, the event-level cost comparison is made only between methods that use the same expanded assignment scope: MinMax with expanded assignment scope and the proposed two-stage assignment with expanded assignment scope.

Figure 5 illustrates the three-target expanded joint assignment event. Target 3 first issues a handover request. During the assignment process, S_{20} is found to have already been committed to Target 2, so Target 2 is added to the expanded joint assignment. A further committed-interceptor overlap occurs when the assignment process involving Target 2 includes S_{13}, which has already been committed to Target 1. As a result, the assignment scope expands from a single calling target to the three-target assignment scope $[T_3, T_2, T_1]$. This example shows that the expanded joint assignment scope prevents previously served targets from being excluded when their committed interceptors are contested by later handover requests.

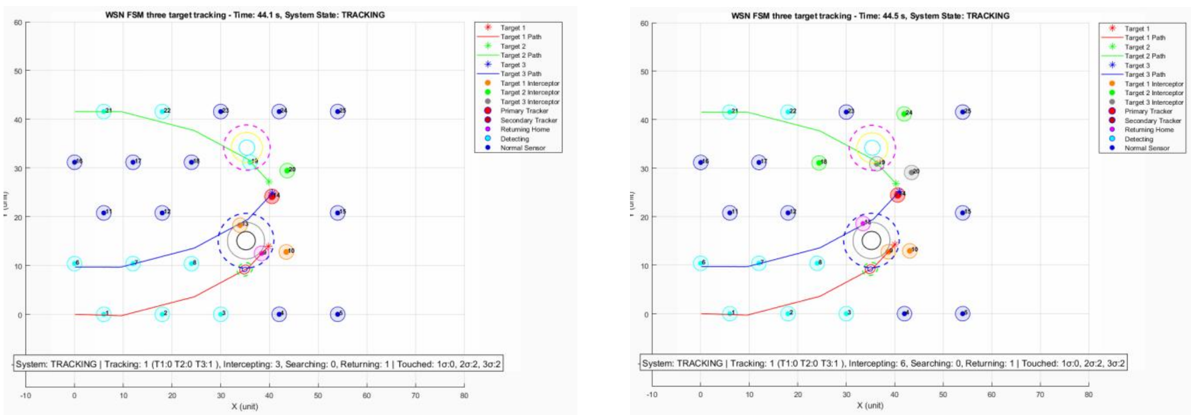

Figure 5: Three-target expanded joint assignment event at $t = 44.2s$.

Once the expanded joint assignment scope is fixed, MinMax and the proposed two-stage assignment differ only in their assignment objective. MinMax prioritizes the worst target-level team cost, while the proposed two-stage objective first maximizes the number of filled interceptor slots and then minimizes the total assignment cost.

Method	Target 3 assignment (team cost)	Target 2 assignment (team cost)	Target 1 assignment (team cost)	Worst team cost	Total Cost
MinMax with expanded joint assignment scope	S_{19}, S_{20} (0.57317)	S_{18}, S_{24} (0.67203)	S_9, S_{10} (0.22667)	0.67203	1.47187
Two-Stage with expanded joint assignment scope	S_{13}, S_{15} (0.64247)	S_{19}, S_{20} (0.46429)	S_9, S_{10} (0.22667)	0.64247	1.33343

Table 3: Assignment comparison for the three-target event.

Table 3 shows that both methods fill all six requested interceptor slots, but the two-stage assignment objective achieves a lower total assignment cost, reducing the cost from 1.47187 to 1.33343. This result supports the event-level objective defined in this report: after the filled-slot maximization step is satisfied, the two-stage method selects the feasible non-overlapping assignment with the lowest total cost.

To complement the event-level comparison, the total tracked target time is also recorded over the full simulation. This metric is reported in target-seconds, where one target-second represents one target being successfully tracked for one second.

Method	T1 tracked time	T2 tracked time	T3 tracked time	Total tracked time ratio
MinMax with expanded joint assignment scope	27.40 / 75.00 s	25.20 / 71.00 s	17.30 / 68.50 s	32.59%
Two-Stage with expanded joint assignment scope	39.00 / 75.00 s	14.70 / 71.00 s	14.80 / 68.50 s	31.94%

Table 4: Full-simulation tracked-target time comparison for the three-target example.

Table 4 provides a complementary observation rather than the primary event-level comparison. In this simulation seed, the two-stage assignment objective produces a slightly lower total tracked-time ratio than the MinMax variant, 31.94% compared with 32.59%. The per-target results also show that the two-stage objective shifts more tracking time toward Target 1, while Target 2 and Target 3 receive less tracked time than under MinMax. This does not contradict the event-level assignment result, because full-simulation tracking time depends on later target motion, subsequent handover timing, future conflicts, and downstream effects of earlier assignment decisions.

Overall, this example shows that the expanded joint assignment scope and assignment-objective selection should be understood separately. The expanded joint assignment scope determines which targets are included in the assignment problem, while the choice between MinMax and the two-stage assignment objective determines how the system prioritizes balanced target-level support versus lower event-level total assignment cost.

6. Conclusion and Future Work

This report studied interceptor assignment conflicts in a mobile WSN for cooperative multi-target tracking. Building on the DES-based proactive handover framework in [1], the report focused on the case where a later handover request contests a sensor that has already been committed as an interceptor for another target. To address this issue, the proposed method expands the joint assignment scope beyond the current calling targets so that any previously served target whose committed interceptor is contested is also included in the assignment problem. The resulting assignment is solved using a two-stage objective that first maximizes the number of filled interceptor slots and then minimizes the total assignment cost. Simulation examples show that the expanded joint assignment prevents previously served targets from being excluded during committed-interceptor conflicts, while the two-stage assignment objective achieves lower event-level total assignment cost than MinMax in the examined three-target event. The results also show that MinMax remains a meaningful fairness-oriented baseline, while the proposed two-stage method prioritizes event-level total assignment cost minimization.

Future work should evaluate the proposed method under more realistic communication and hardware constraints. The current simulation assumes ideal communication, but physical deployment would require measuring bid propagation delay, packet loss, commitment confirmation time, and the possibility of inconsistent assignment reconstruction among sensors. In addition, the simulation results suggest that lower event-level assignment cost does not necessarily guarantee better aggregate tracked-target time in every seed. Future work should therefore investigate adaptive objective selection, where the system can switch between MinMax and the two-stage objective depending on target priority, uncertainty level, fairness requirements, or long-horizon tracking performance. More simulation seeds and more diverse multi-target trajectories should also be tested to evaluate the robustness of the expanded joint assignment under different conflict patterns.

7. References

[1] N. Najafizadeh Sari, Y. Sang, G. Nejat, and B. Benhabib, "Reconfigurable Wireless Sensor Network Coordination for Simultaneous Multi-Target Tracking"

8. Appendix:

- example_V47_Yeqi.m — main simulation script for the V47 expanded joint assignment experiment.
- example2-expanded-group-two-stage.log — log file for the expanded joint assignment experiment using the two-stage assignment objective.
- example2- expanded-group-minmax.log — log file for the expanded joint assignment experiment using the MinMax assignment objective.